

# Sean Luka Girgis

Senior Data Engineer / Time-Series Forecasting Engineer

214-315-2190 | seanlgirgis@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## Professional Summary

Senior Data Engineer with 20+ years of enterprise IT experience across Python data pipelines, telemetry analytics, time-series forecasting, capacity planning, operational reporting, and performance engineering. Strong background building scalable data workflows, forecasting datasets, anomaly/risk detection patterns, stakeholder dashboards, and governed reporting systems using Python, SQL, PySpark/Spark, Pandas, Prophet, scikit-learn, Oracle, AWS data platform patterns, and production monitoring data. Experienced translating complex infrastructure and machine-like telemetry into actionable insights, forecasts, and recommendations for technical and non-technical stakeholders.

## Professional Experience

### Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas ETL pipelines ingesting P95 telemetry from 6,000+ infrastructure endpoints, replacing manual processes with repeatable, validated data workflows.
- Developed time-series forecasting models using Prophet and scikit-learn to predict capacity bottlenecks months ahead and support proactive operational planning.
- Processed large-scale infrastructure telemetry using Python, Pandas, SQL, and PySpark/Spark patterns to support forecasting, anomaly-style risk detection, and executive reporting.
- Designed optimized SQL/Oracle schemas and reporting structures for historical telemetry retention, trend analysis, seasonal forecasting, and operational decision support.
- Integrated BMC TrueSight/TSCO, CA APM, and AppDynamics feeds into unified reporting datasets with reconciliation, validation, and data quality checks.
- Built hybrid AWS/Oracle data platform patterns using S3 landing zones, AWS Glue ETL concepts, Redshift analytics workloads, and Oracle storage for forecasting and reporting.
- Created dashboards and analytical outputs using Streamlit, matplotlib, seaborn, and plotly to communicate capacity risk, utilization trends, performance patterns, and recommendations to stakeholders.
- Documented data definitions, feature preparation logic, model assumptions, transformation rules, validation checks, and operational reporting workflows to support continuity and governance.

### Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, strengthening visibility into machine and application health signals.
- Led the FAST project, mining real-user and synthetic performance metrics to surface optimization opportunities and provide actionable recommendations to engineering teams.
- Implemented End-User Monitoring and transaction tracing across web and mobile systems to improve visibility into user journeys, API behavior, and application performance.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM telemetry, and infrastructure metrics.

- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and performance data requirements.

#### **SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)**

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application health and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and replace manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure KPIs.
- Delivered technical documentation, onboarding guidance, and client training for monitoring standards, deployment processes, and production troubleshooting.

#### **Performance Test Engineer at AT&T (2010-08 – 2011-07)**

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used Java/J2EE, JMX monitoring, Wily Introscope, and thread-dump automation practices to provide real-time visibility during performance validation.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

#### **Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)**

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness across large transaction datasets during phased cutover.
- Automated schema conversion and data validation workflows to protect data integrity across relational database migration paths.

#### **Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)**

- Developed high-availability telecom billing interfaces using C, C++, Pro\*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and Unix environments.
- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained reporting, invoicing, settlement, and integration components across full billing lifecycle systems for large-scale telecom operations.
- Worked with enterprise integration and messaging patterns including MQ Series, XML-based workflows, WebLogic/WebSphere, and Unix automation.

## Education

### **High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science**

Humber College, Toronto, Canada

### **Bachelor of Science in Civil Engineering**

Zagazig University, Egypt / Cairo, Egypt

## Skills

**Languages:** Python, SQL, PySpark, PL/SQL, C++, C, Java, Perl, Korn Shell, XML, JSON

## Flagship Projects

### **HorizonScale — AI Capacity Forecasting Engine**

Designed a forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into capacity-risk predictions and operational decision support.

**Technologies:** Python, PySpark, Prophet, scikit-learn, Streamlit, Time-Series Forecasting, Telemetry Data

### **Performance Telemetry and Capacity Reporting Platform**

Built Python and SQL-driven reporting workflows to turn monitoring and infrastructure telemetry into dashboards, validated datasets, executive reports, and trend analysis.

**Technologies:** Python, SQL, Oracle, BMC TrueSight, AppDynamics, CA APM, Streamlit

### **Serverless Lakehouse Platform (AWS)**

Designed a serverless data platform pattern using S3, Glue, Athena, Redshift, Bedrock, PySpark, Parquet, and Snappy compression for cloud-scale ingestion, transformation, query, and AI-enabled access.

**Technologies:** AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet, Bedrock

### **AI-Powered Job Search Pipeline**

Built an agentic Python pipeline for job intake, semantic duplicate detection, LLM scoring, JSON artifact generation, quality gates, and tracking using FAISS, sentence-transformers, Grok API, and Claude Code workflows.

**Technologies:** Python, FAISS, Sentence Transformers, LLM APIs, RAG, Agentic Workflows, JSON Contracts